frontiers
in Astronomy and
Space Sciences

TYPE Original Research
PUBLISHED 29 July 2026
DOI 10.3389/fspas.2026.1848834

Check for updates

OPEN ACCESS

EDITED BY
Guozhu Li,
Institute of Geology and
Geophysics, China

REVIEWED BY
Biqiang Zhao,
Institute of Geology and
Geophysics, China
Changrong Ye,
Chongqing University of Posts and
Telecommunications, China

*CORRESPONDENCE
J. A. Aricoche,
jaa387@cornell.edu

RECEIVED 06 April 2026
REVISED 11 June 2026
ACCEPTED 02 July 2026
PUBLISHED 29 July 2026
CORRECTED 06 August 2026

CITATION
Aricoche JA, Hysell DL and Morton YJ (2026) Probing the main ionospheric trough with an Alaskan HF beacon network.
Front. Astron. Space Sci. 13:1848834.
doi: 10.3389/fspas.2026.1848834

COPYRIGHT
© 2026 Aricoche, Hysell and Morton. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

Probing the main ionospheric trough with an Alaskan HF beacon network

J. A. Aricoche[1]*, D. L. Hysell[2] and Y. J. Morton[3]

[1]Electrical and Computer Engineering, Cornell University, Ithaca, NY, United States, [2]Earth and Atmospheric Sciences, Cornell University, Ithaca, NY, United States, [3]Aerospace Engineering Sciences, University of Colorado Boulder, Boulder, CO, United States

An HF beacon network has been deployed in eastern Alaska to estimate plasma number densities in the ionospheric volume overhead. The network includes three dual-frequency CW transmitters employing pseudorandom noise (PRN) codes and three receivers. The network spans latitudes 62–68° north geographic. Observables from the network include the signal amplitude, Doppler shift, and group delay of all the ray paths joining the transmitters and receivers with a 1 min Cadence. The data are interpreted using a forward model based on geometric optics and an inverse method which solves for the parametrized ionospheric plasma number density in the volume overhead as an optimization problem. Sensitivity analysis is incorporated through automatic differentiation. Additional information from regional GNSS receivers, ionosondes, and incoherent scatter radars can be included in the inverse method. Range and Doppler spreading of the HF signals can furthermore be interpreted in terms of ionospheric turbulence and irregularities along the ray paths. This paper describes the methodology involved and compares electron density estimates with measurements from the Poker Flat Incoherent Scatter Radar (PFISR) which were withheld from the inverse methodology for validation.

KEYWORDS
Doppler spreading, geometric optics, HF beacon network, main trough, optimization, range spreading

1 Introduction

Whereas experimentalists have numerous means at their disposal for measuring electron number densities along spatial trajectories, as profiles, or as path-integrated quantities, few experimental modalities measure electron number densities throughout a volume. Incoherent scatter radars (ISRs) have this capability. However, ISRs have limitations for probing large volumes with fine spatial (range and bearing) and temporal resolution arising from stationarity constraints on the incoherent scatter measurement. The tradeoffs are complicated and depend on the characteristics of the radar in question. ISRs are, in any case, expensive, in high demand, and in short supply.

Alternatives for volumetric electron density measurements are networks of radio instruments. In the framework of geometric optics, each ray conveys information about the integrated electron number density, leaving the volumetric number density to be estimated using some form of tomography. Incorporating high-frequency (HF) signals can be advantageous since this invites the utilization of ground-based instruments such as beacons and sounders. Another advantage of HF signals is that their group delay, Doppler shift, and amplitude yield independent information about the intervening ionosphere due to

dispersion and nondeviative absorption at HF. Some spacecraft-based measurements (i.e., GNSS signals) are nonetheless required if electron number densities are to be estimated in the topside. VLF signals, meanwhile, can provide additional information about the *D* region. The measurement cadence for radio measurements can be relatively high. Spatial resolution, meanwhile, will be controlled in large part by the density of deployed instruments.

An HF beacon network comprising three dual-frequency transmitters and three receivers was deployed in Alaska to investigate the viability of volumetric electron density reconstruction over a wide area. The effort follows a similar one conducted recently in Peru (Aricoche and Hysell, 2024). The focus of the previous study was on the evolution of the equatorial ionosphere during the passage of the prereversal enhancement and the subsequent formation of large-scale plasma depletions and irregularities associated with equatorial spread *F* (ESF). Here, the interest is in postsunset ionospheric structuring around the main ionospheric trough (MIT).

The MIT was first discovered and described in the pioneering work of Muldrew (1965), with earlier evidence from topside HF sounding reported by Knecht and Zandt (1963). The MIT forms from a complex interaction between the sub-auroral ionosphere and the magnetosphere and is seen as a narrow latitudinal region of depleted plasma density in the *F* region, generally corresponding to the nighttime equatorward boundary of the auroral oval (Rodger, 2008). It is usually divided into three sections: the equatorward wall, the trough minimum, and the poleward wall. Modeling and data assimilation studies first demonstrated that sharp gradients in high latitude convection can bring together plasma parcels with very different precipitation and heating histories, thereby carving out a trough-shaped depletion (Quegan et al., 1989). Subsequent multi-satellite observations confirm that flows immediately poleward or equatorward of one another often traverse markedly different regions before converging near sub-auroral latitudes, thus sustaining the density deficit characteristic of the MIT (Aa et al., 2020a). Volumetric estimates of plasma number density in the vicinity of the MIT would be useful for distinguishing the formation processes at work.

The HF signals at the center of this work provide information not only about the background regional electron number density but also about ionospheric irregularities which cause HF scintillations. Scintillation models for HF sky-wave signals can be classified into three groups as outlined by Ding et al. (2021). Geometric optics phase models consider the ionosphere as a slightly perturbed refractive medium and utilize a first-order Taylor expansion to assess phase perturbations along the ray path (Coleman, 1996; Ravan et al., 2012). While these models can replicate spatial and temporal phase variations, they neglect diffraction, making them inadequate for predicting amplitude fluctuations. The second category comprises physical models based on the extended Rytov approximation which incorporates diffraction within an arbitrary three-dimensional inhomogeneous background medium and can simulate both phase and amplitude scintillation provided the fluctuations are not too strong (Gherm et al., 2005; Zernov et al., 2019). The final category includes multiphase-screen models (MPS) (Wagen and Yeh, 1989; Nickisch et al., 2012; Ding et al., 2021) that calculate random phase changes and diffraction effects. However, MPS techniques often depend on specific parameterizations regarding screen separation which can be unmanageable for general modeling purposes. Conversely, Coleman's geometric optics clutter model within the geometric optics category illustrates that broadening and spreading can be predicted directly from ray paths as an alternative approach.

This paper presents a technique for volumetric electron density estimation using HF beacons supported by other radio instruments, here specifically in the main ionospheric trough. It further explores the impacts of irregularities in the geometric optics paradigm by quantifying Doppler and range spreading, both posing hazards to HF communication and radar systems. Below, a specific methodology is described. Following that, a dataset is analyzed as a case study. Finally, results of the exercise and implications for further work are evaluated.

2 Methodology

The HF beacon network in Alaska operates at 2.9 and 3.4 MHz and employs distinct pseudorandom noise codes (PRNs) as described recently by Vierinen et al. (2016) and Tineo (2026) for application with low-power bistatic sounders. The codes have 10μs bit widths and repetition times of 0.1 s, implying a compression ratio of 10,000. Despite originating from low-power sources (10 W) and low-gain antennas, the first-hop sky-wave signals of interest here typically have signal-to-noise ratios in excess of 20 dB. When decoded with a least-squares filter, the computed ambiguity functions of the PRN codes are essentially ideal, with sidelobes down by a factor of more than 150 dB. Crosstalk between different codes is suppressed by 30–40 dB.

Beacon stations are synchronized using GPS-disciplined oscillators (GPSDOs) with an accuracy of ±50 ns. Synchronization has been tested and meets the requirement for the signal bandwidth in use here. The software-defined transcievers (SDRs) themselves have 87–88 dBc dynamic range. We do not use automatic gain control but merely set the receiver gain such that the background noise is strong enough to toggle several bits. The receive sites are remote and quiet, and signals are not observed to saturate.

The 2.9 and 3.4 MHz bistatic skywave signals representing the bulk of the data in this paper appear as compact echo clusters in Range-Doppler space. Multi-hop signals are common in the data, but multipath signals are not observed, and only the low-path signal is seen. Most often, the two magneto-ionic modes are not distinct and separated, and we interpret the received signal as representing the ordinary mode. When the magneto-ionic modes are distinct, we track the stronger of the two and interpret that as the ordinary mode. Note that the analysis described below is general and could accommodate multipath and multimode signals. In the general case, multiple rays could simply be assigned to multiple signal components without contradiction. To reiterate, the absence of multi-mode, multi-path, and distinct high and low paths is not an assumption but rather an observation.

The channel capacity of the ionosphere at HF was considered by Wagner and Goldstein (1985) and references therein who considered the propagation of HF signals with bandwidths up to 1 MHz. They defined the channel capacity as $B = (d\tau/df)^{-1/2}$ where τ is the group propagation delay time and f is the carrier frequency. The sensitivity of the propagation delay time to the carrier

TABLE 1 Transmitter and Receiver Locations in Alaska in geodetic coordinates. The first two stations are transmitters, and the last three are receivers. Note that a third transmitter was also deployed to Homer, Alaska. However, signals from that most distant station were received only sporadically and so will not be considered here.

Station	Latitude (north)	Longitude (west)	Altitude (masl)
Gakona	62.390	−145.150	450
Poker flat	65.122	−147.478	210
Toolik	68.630	−149.600	720
Eagle	64.780	−141.200	260
Trapper creek	62.320	−150.230	110

frequency is calculated by the raytracing algorithm described below for the ray paths modeled to reproduce the HF beacon observations. The algorithm reports that most ray paths have sensitivities between 15 and 50μs/100 kHz signifying channel capacities between about 50 and 80 kHz. This is only a little smaller than the bandwidth of the PRN-coded signals, implying minimal range broadening due to signal dispersion at the frequencies in question. We therefore do not consider dispersion, multi-mode, and multipath effects on the range-spreading of the HF signals considered here, recognizing that these factors could be important in the general case.

Broadband inverted-V antennas are used for transmission, and loop antennas are used for reception. Receivers sample at a rate of 1 M/s, and the effective sample rate is reduced to 100 kS/s through decimation, yielding an equivalent monostatic range resolution of 1.5 km. Each range gate is sampled every 100 m and coherently processed for 10 s. The signal is then detected. Finally, the spectra are incoherently averaged for 1 min And then analyzed for group delay, Doppler shift, and amplitude. Additional technical details were given by Hysell et al. (2018) for a similar network developed in Perú. The locations of the transmitters and receivers are listed in Table 1, and a map of HF beacon stations in Alaska is provided in Figure 1.

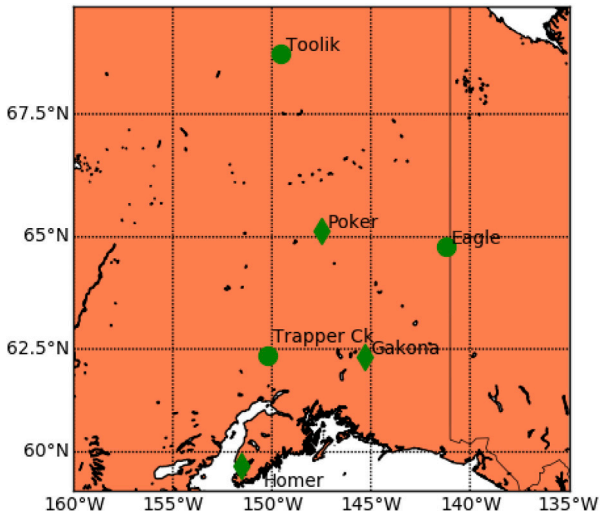


FIGURE 1
Map of HF transmitters (diamonds) and receivers (circles) deployed in Alaska. The transmitter in Homer is not used in this analysis. The GNSS receiver, ionosonde, and PFISR radar are also located at Poker Flat.

2.1 HF beacon optimization problem

The methodology for inferring volumetric electron number density estimates from the observables was described in detail by Hysell et al. (2021) and Aricoche and Hysell, (2024). It employs the equations of geometric optics posed in geocentric coordinates in the forward model (Jones and Stephenson, 1975). The Hamiltonian is based on the Appleton-Hartree equation and includes the effects of the geomagnetic field and electron-neutral collisions. The equations are solved as an initial-value problem using numerical quadrature with a stiff adaptive method. Rays are traced from the transmitters through a parametrized ionosphere to the ground. The initial bearings of the rays are tuned until the ground intercept points match the respective receiver locations. At this point in the process, all the observables are predicted.

The parametrization coefficients describe a generalized five-parameter Chapman function in the vertical which specify a peak F-region electron density, peak altitude, bottomside scale height, topside scale height, and the density of an added E layer with a Gaussian profile. These coefficients are themselves expanded in the horizontal using biquintic B-spline interpolation. The horizontal grid is nominally 15×15, implying 1,125 coefficients that must be set. These coefficients are adjusted iteratively as part of an optimization problem (Powell, 1964) that minimizes the discrepancies between the predicted and measured observables in the L2 sense. The minimization method is Levenberg Marquardt with regularization that penalizes the local curvature everywhere. The objective function includes weights based on confidence estimators for all the observables. Optimization is carried out at 1-min intervals, and the initial guess for each interval is the solution to the previous interval.

The geometric optics equations in use are complicated, owing to the use of spherical coordinates and to the many coordinate transformations involved that relate the Hamiltonian to the index of refraction to the magneto-ionic parameters X, Y_l, Y_t, and Z to the geomagnetic field, where $X \equiv \omega_{pe}^2/\omega^2$, Y_l and Y_t denote the longitudinal and transverse components of $Y \equiv \Omega_e/\omega$, respectively, and $Z \equiv \nu_{en}/\omega$. Together with the electron-neutral collision frequency and the electron number density, these parameters determine the model coefficients. Note that the equations of geometric optics require not only the Hamiltonian but also its gradients with respect to the ray state vector $(\mathbf{k}, \mathbf{x}, \omega, t)$, where

k is the scattering wavenumber, ω is the Doppler frequency and t is the signal time of flight.

Sensitivity analysis is required both to adjust the initial ray bearings in the forward model and to calculate the amplitudes of the received rays which undergo deviative absorption. While sensitivity analysis can be performed using either variational (forward) or adjoint (reverse) methods, this would be very tedious, as not only the gradient vector of the Hamiltonian but also its Jacobian and Hessian matrices would be required (Coleman, 2011; Baumgarten et al., 2021). For this study, we instead implemented automatic differentiation (AD) (Moses et al., 2021; Moses et al., 2022). AD employs computational tools that effectively rewrite the source code, applying knowledge about the derivatives of elementary functions as well as the chain rule to the program trace. In reverse mode AD, the computed terminal ray coordinates are automatically accompanied by the sensitivities of the coordinates to the control parameters (the initial ray bearings in this case). This is done to machine precision without computational penalty compared to alternative methods. AD simplifies code development drastically and reduces the likelihood of coding errors. AD is applicable to algorithms along with explicit formula computations and yields a method that is highly adaptable. While AD is presently only applied in the raytracing component of the forward problem, it could equally well be applied in the solution of the overall inverse problem. This would simplify error propagation this is complicated and which has not yet been implemented with our algorithm.

Note that the optimization problem described above can easily accommodate generalizations to its objective function. The method used here incorporates GNSS line-of-sight total electron content (TEC) measurements along with foF2, HmF2, and layer width estimates from the ionosonde at Poker Flat. ISR measurements could equally well be incorporated but were withheld in this study for later validation. The cost function is a weighted sum of all residuals across all measurements, together with regularization terms.

2.2 Range-Doppler spectrogram analysis

Data from each receiver are processed for each frequency and PRN code separately, yielding Range-Doppler spectrograms for every timestep. The group delay, Doppler shift, and amplitude of the first-hop signal are then determined and tabulated.

The power spectrum is characterized through a two-dimensional Gaussian form as an expedient way of capturing the mean and spread, without implying that the spectrum itself has a gaussian shape. The prominent peaks in each spectrogram correspond to the maxima of the Gaussian, while the standard deviation indicates the Doppler spread in Hz and the group delay in km. The spectral resolution is 0.1 Hz and 1.5 km. To extract the parameters defining the Gaussian shapes, we utilize an algorithm that finds two dimensional Gaussians in pixelated data. From the obtained parameters, the Doppler shift, group delay, and amplitude are estimated. The inversion is executed on the unfiltered and clipped SNR spectra based on the initial guesses.

The initial guesses for this process are based on the maxima identified in the filtered spectra. The filtering process applied to the spectrograms is illustrated in Figure 2. First, the SNR of the power spectrum are taken. Then, a mean filter is employed which smooths images by reducing intensity variations across adjacent pixels. This is followed by applying a Boolean mask based on a percentile threshold. This step was performed to facilitate the labeling of binary images. Finally, we use a connected component labeling algorithm. This algorithm assigns labels to pixels so that adjacent pixels that share similar features receive the same label. Developed initially by Fiorio and Gustedt, (1996) and later optimized by Wu et al. (2005), connected component labeling is effective to ensure that all pixels with the same label are spatially contiguous, implying that each area forms a solid object. This is a capability that generic clustering algorithms might not always provide and is crucial for maximizing the quality of the estimates obtained from the spectrograms, especially under noisy conditions. This processing is suitable because the accuracy of the inversion relies on the initial guesses. It can be seen in the first row of Figure 2 that, due to the nature of some spectra, a simple peak identification algorithm would be insufficient. By applying the filters, the spectrum is identified, and the peaks are located with good precision.

The second row of Figure 2 depicts how both the first and second hop signal are identified. Locating the maximum peaks in the properly filtered signals is essential for establishing the boundaries in which the inversion will be applied, specifically around the first hop of the F layer. In Figure 3, two distinct regions where the inversion method is utilized are highlighted (One region extends from 200 km to 350 km, while the other extends from approximately 300 km–450 km). These specific regions vary between ray paths and frequencies depending on where the maxima are located. We encounter different types of spectra some showing no spreading at all, and others exhibiting range spread, Doppler spread, or a combination of both. For example, the left side of Figure 3 shows a spectrogram with significant Doppler spread whereas the right side shows a spectrum with significant range and Doppler spread.

3 Case study analysis

We consider measurements from Alaska made on Nov. 8, 2023. This was a geomagnetically quiet interval when the corrected solar flux index (10.7 cm) was 143.1. We concentrate on universal times between about 0,200 and 0340 UT, corresponding approximately to local sunset, the signals received by the HF network were strongest because the ionospheric absorption had decreased while the F layer was only beginning to recombine. Usable HF data were not recorded outside this time interval.

The analysis incorporates data from a GNSS receiver at Poker Flat. We utilize total electron content (TEC) measurements from a fixed number of satellites. In particular, we only use measurements from the three PRNs with the smallest zenith angles at any given time. The measurements used here are depicted in Figure 4. These measurements show a gradual decrease in postsunset electron density but also could suggest steep horizontal gradients as the measurements from different PRNs differ markedly at times.

The analysis also incorporates data from the ionosonde at Poker Flat. Specifically, we consider the autoscaled parameters foF2, HmF2, and a width parameter which is indicative of the bottomside ionospheric scale height. Measurements for Nov. 8, 2023, are shown in Figure 5. Note that the experimental cadence for these parameters is 15 min. We utilize Gaussian process regression

10.3389/fspas.2026.1848834

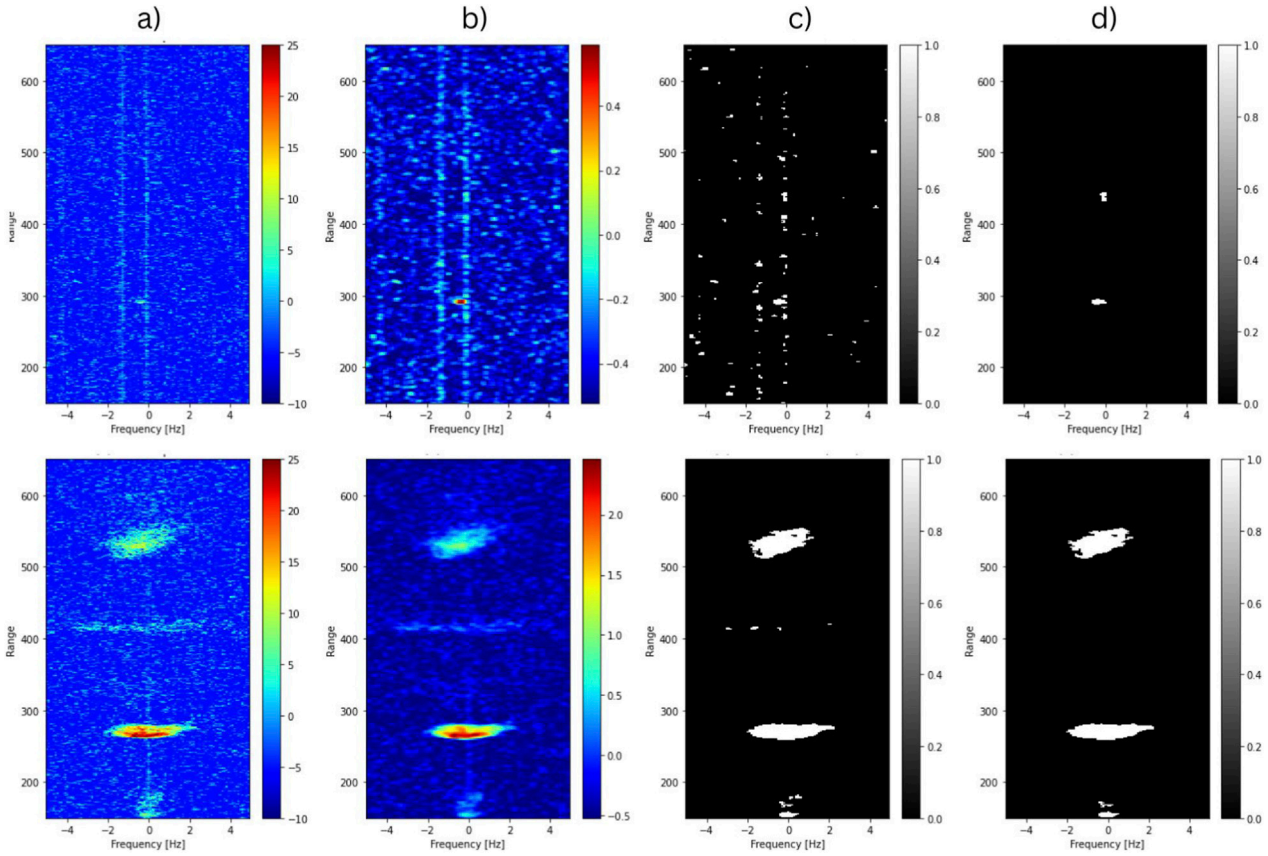


FIGURE 2
(a) SNR of the spectrum, **(b)** Mean filter, **(c)** Percentile mask, **(d)** Connected component labeled filter.

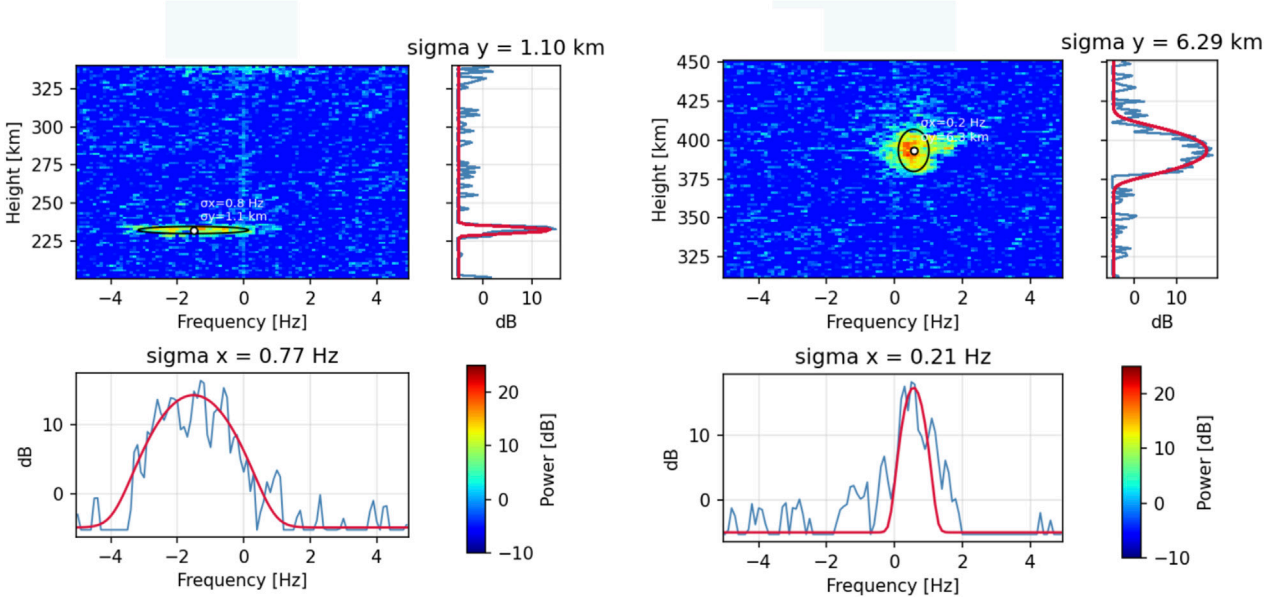


FIGURE 3
Two different SNR spectrograms along with the standard deviations obtained for each spectrum: sigma x for Doppler spread and sigma y for range spread. The blue lines indicate the SNR in decibels (dB) along cuts through the peak. The red curves illustrate the Gaussian model reconstruction through the peak.

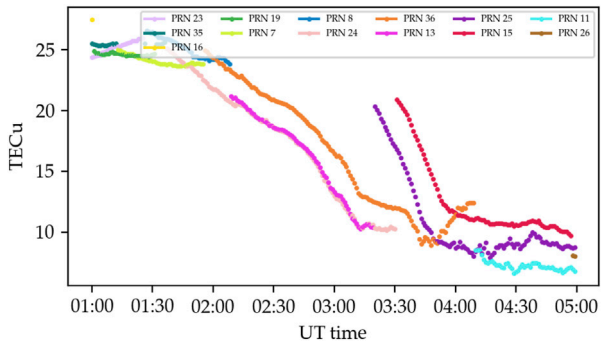


FIGURE 4
GNSS measurements from Poker Flat capturing the three GPS PRNs with the smallest zenith angle at any given time.

to interpolate down to the cadence of our HF data processing algorithm. The curves in Figure 5 imply a post-sunset *F* layer that is gradually decreasing in density and increasing in height while arguably exhibiting periodic fluctuations. The layer seems to have undergone a relatively abrupt change at about 0300 UT when all three measured parameters decreased simultaneously.

HF data for the postsunset hours of Nov. 8, 2023, are shown in Figure 6. The lower-left panel shows the group delay for all the ray paths. Our convention is to plot the product of the group delay time and half the speed of light, the equivalent monostatic radar range. The group delays are mainly found to increase with time, suggesting gradual *F*-layer ascent, decay, or both. Some of the paths also exhibit small oscillations. These oscillations are more distinct in the 2.9 MHz data than the 3.4 MHz data despite the two frequencies being relatively close.

The lower-right panel of Figure 6 shows the accumulated carrier phase for the same ray paths. Whereas the Doppler shift is the actual observable and the parameter used in the subsequent analysis, it is expedient to plot accumulated carrier phase here instead for illustrative purposes. The accumulated carrier phase is calculated by integrating the Doppler shift numerically. An arbitrary constant is added to each curve to make it align vertically with the corresponding group delay curve, facilitating easier comparison. In this case, the comparison reveals that the former and the latter exhibit similar behavior overall while differing quantitatively in the details. This is to be expected, as the two observables are related to the underlying plasma state variables through different transfer functions at HF frequencies. The two observables therefore provide distinct information for the inverse problem.

Finally, the upper panel of Figure 6 shows the relative amplitudes of the received signals. All of the signals exhibit approximately 20 dB of fading over the data accumulation period. The fading is deeper and more rapid than that typically observed in the equatorial zone, even during intervals of equatorial spread *F* (Aricoche and Hysell, 2024). Some of this can be captured by the inversion algorithm which accounts for deviative and nondeviative absorption. The most rapid fading is beyond the ability of the algorithm to capture.

Electron density estimates derived from the aforementioned HF, GPS, and ionosonde data are depicted in Figure 7 in the form of isodensity surfaces. Representative snapshots are shown. intervals beginning at 0200 UT (Solutions are available at 1-min intervals.) The salient features of the electron density reconstruction are as follows. First, there is a broad longitudinal gradient in electron density that remains detectable throughout most of the analysis period but diminishes over time. This can be attributed to sunset and the variation of solar zenith angle across the solution domain. Second, there is also a latitudinal gradient in electron number density that is steeper than the longitudinal gradient.

Two dimensional slices of the electron densities estimates are included in Figure 8 at a fixed longitude of -147°. Amidst considerable variability, we may expect the main ionospheric trough to be deepest at about 65° magnetic latitude (or about 64° geographic latitude) with a width between 6–7° in early November (Aa et al., 2020b). Early in the event in question, the latitudinal gradient appears to delineate the poleward boundary of the main trough. As time progresses, the latitudinal gradient varies and weakens as the poleward boundary appears eventually to move northward and out of the domain.

After about 0300 UT, the trough minimum becomes evident at higher latitudes. By the end of the interval shown, the equatorward boundary of the trough emerges at the lowest latitudes in the domain. The morphology of the density surfaces shown in Figure 7 pivots at about 0300 UT as the trough minimum passes through the latitudinal center of the solution space. During the course of less than 2 hours, the defining features and evolution of the MIT could therefore be identified in the HF beacon network diagnostic.

While the HF data analysis method can readily incorporate electron density profiles from ISRs, a substantial data gap in the Poker Flat Incoherent Scatter Radar (PFISR) data during our experiment prevented direct assimilation here. Instead, we use the PFISR data available for validation. Figure 9 shows PFISR electron density profile measurements made with a vertically-looking beam during the experiments taken every 3 min. These are plotted alongside the results from the HF analysis method evaluated in the vicinity of Poker Flat. Error bars on the latter were calculated via a Monte Carlo sensitivity analysis which considered uncertainties in the HF and GNSS data which have propagated through to the model output. Specifically, GNSS standard deviations were calculated using the method of Morton et al. (2020) while allowing for background biases at the level of 1.5 TECu consistent with the findings of Coster et al. (2013) and Bourne, (2016). The error bars, which reflect sensitivity to small fluctuations in the driving measurements, are most often very small and always highly correlated in range. They do not fully reflect uncertainty, neglecting biases introduced by regularization. Regularization is essential in underdetermined inverse problems such as this one, but the resulting biases are difficult to quantify (e.g., Hansen, 2010). The most efficacious way to gauge bias is through validation.

Overall, the HF profiles agree reasonably closely with the PFISR measurements. Note that the fluctuations in the latter are mainly stochastic and arise from a finite number of independent samples contributing to the radar profile estimates in each timestep and beam. Beyond that, the *F* peak density and height and scale heights of the topside and bottomside appear to have been reproduced reasonably accurately by the HF analysis method in

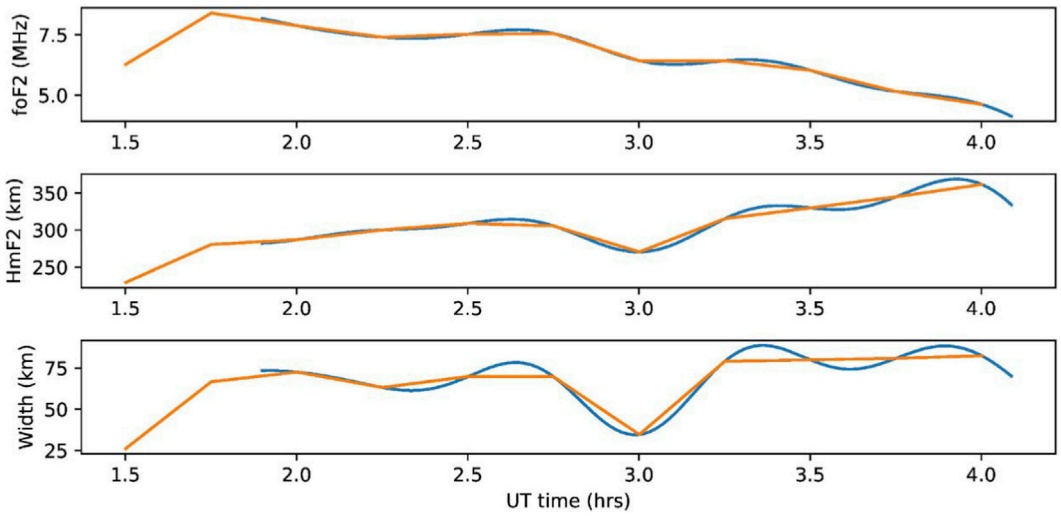


FIGURE 5
Parameters from the ionosonde deployed at Poker Flat determined by autoscaling. Results of interpolation via Gaussian Process Regression are depicted by the blue curves.

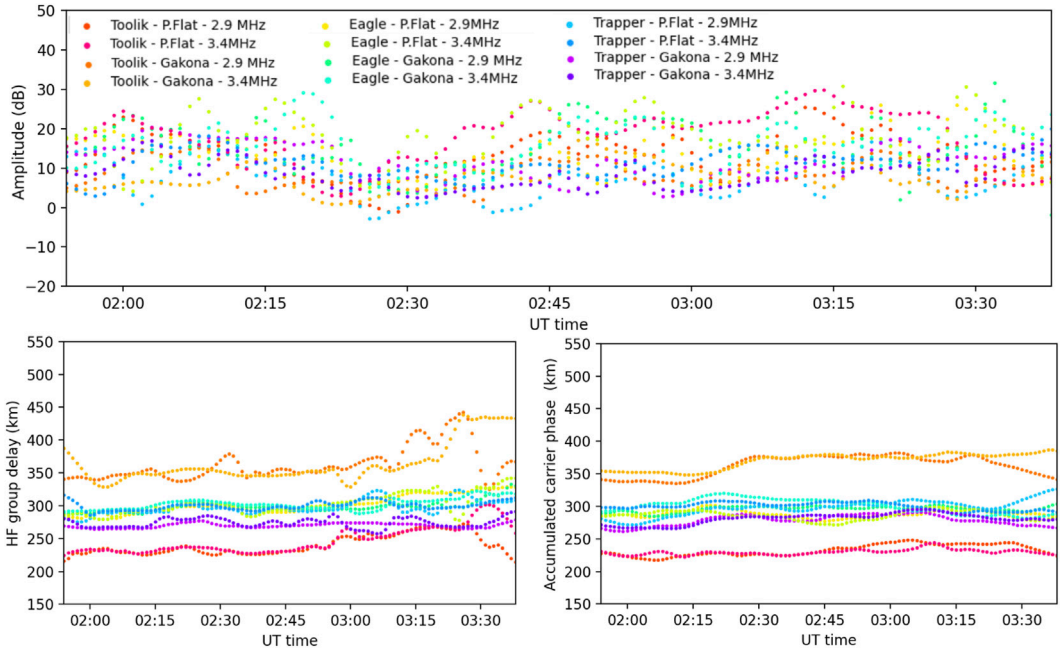


FIGURE 6
Summary of HF beacon data for Nov. 8, 2023 (Upper panel) Relative signal power in dB (Lower left panel) Group delay plotted as equivalent monostatic radar range in km (Lower right panel) Accumulated carrier phase derived from Doppler shifts and plotted as equivalent monostatic radar range in km. The two frequencies for each link are shown: 2.9 MHz and 3.4 MHz.

terms of the confidence intervals shown. As the HF analysis can only yield profiles shaped like Chapman functions, there are inevitable discrepancies. Notably, the PFISR data exhibit local density maxima in the valley region below the *F* peak that cannot be reproduced by the HF method without some further generalization. Discrepancies in the bottomside furthermore imply discrepancies in the topside where the scale heights are effectively set to make the modeled and measured TEC agree. The accuracy of the topside scale heights therefore depends on model profile accuracy at all altitudes. Finally, the validation dataset is admittedly small, reflecting the logistical difficulties involved in acquiring datasets requiring multiple instruments to be functioning in suitable modes simultaneously when ionospheric conditions are favorable and interference is manageable.

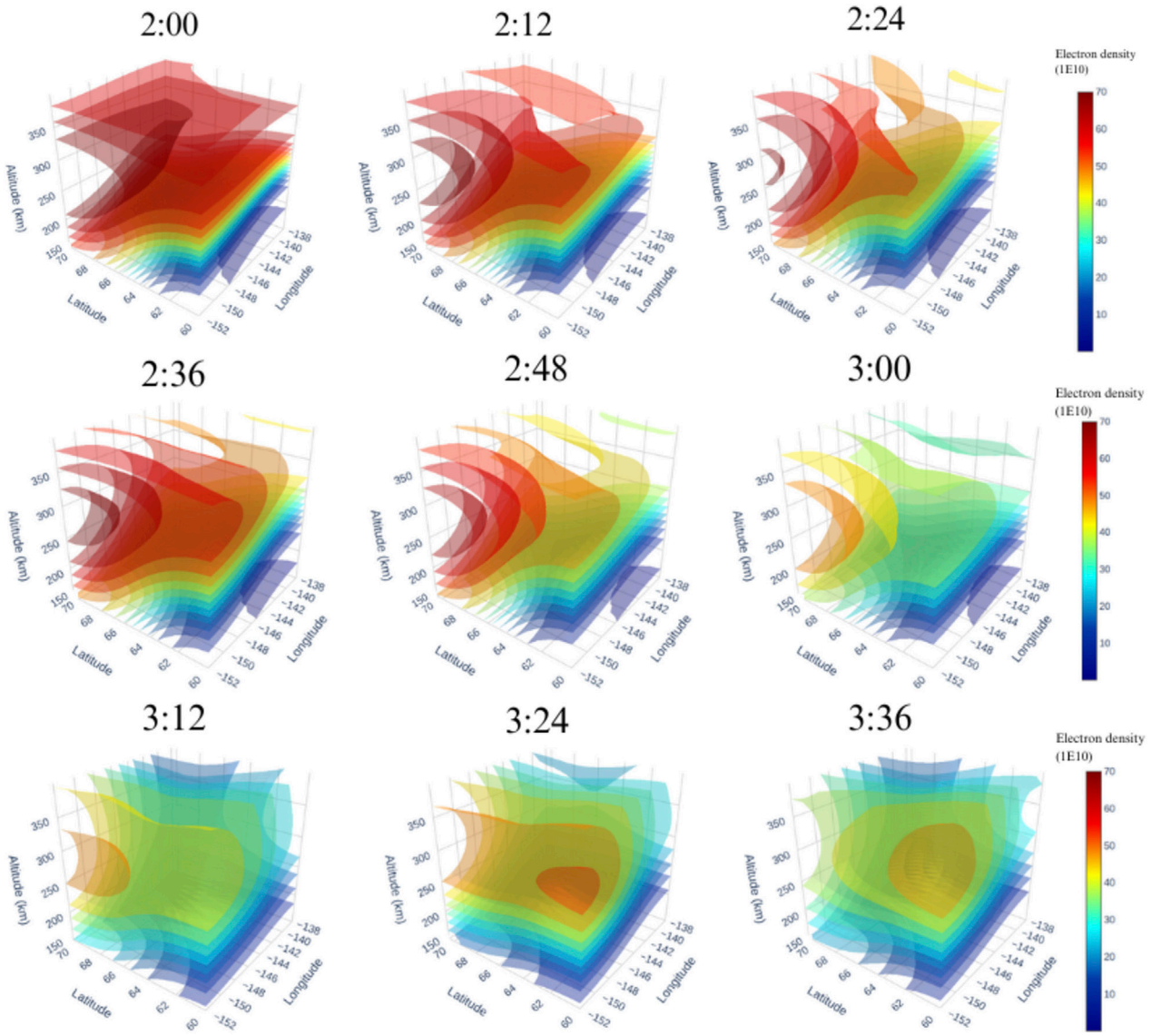


FIGURE 7
Isodensity curves representing the results of the HF beacon data analysis at selected UT times. The colors shown span plasma densities between 1.0×10^9 (violet) and 7.0×10^{11} (red)m^{-3}.

4 Spectral analysis

The range delay and Doppler frequency spreading of the HF signals were estimated and illustrated in Figure 10 for the case study event. "Width" here refers to full width at half maximum (FWHM) which is calculated using the standard deviation from the Gaussian model presented. The figure sorts the measurements by carrier frequency and UT. Points are plotted only when the Gaussian fitting results met a certain confidence threshold. Absolute minima in the plotted points reflect limits imposed by the range and frequency resolution of the transmit signal and subsequent processing.

The instabilities at work in the region of interest are generally accepted to be the gradient drift or $\mathbf{E} \times \mathbf{B}$ instability and the closely-related current convective instability (e.g., Kelley, 1989). The supplementary material reviews a framework for interpreting the spreading of HF clutter caused by ionospheric irregularities and turbulence derived by Coleman, (1996) and extended to quantify both range and Doppler spreading. Evaluating the formalism requires a specification of the wavenumber spectrum of the irregularities which we do not possess, although we can follow Coleman and consider the power-law reference spectrum suggested by Dana and Wittwer, (1991) for preliminary comparison purposes. In summary, the formalism says that range and frequency spreading require ionospheric turbulence, that frequency spreading also requires convection, that frequency spreading is more sensitive to turbulence at relatively smaller spatial scales compared with range spreading, and that spreading should decrease with increasing carrier frequency, range spreading especially.

Figure 10 indicates that range and frequency spreading was generally more prevalent in the second half of the event (after about

10.3389/fspas.2026.1848834

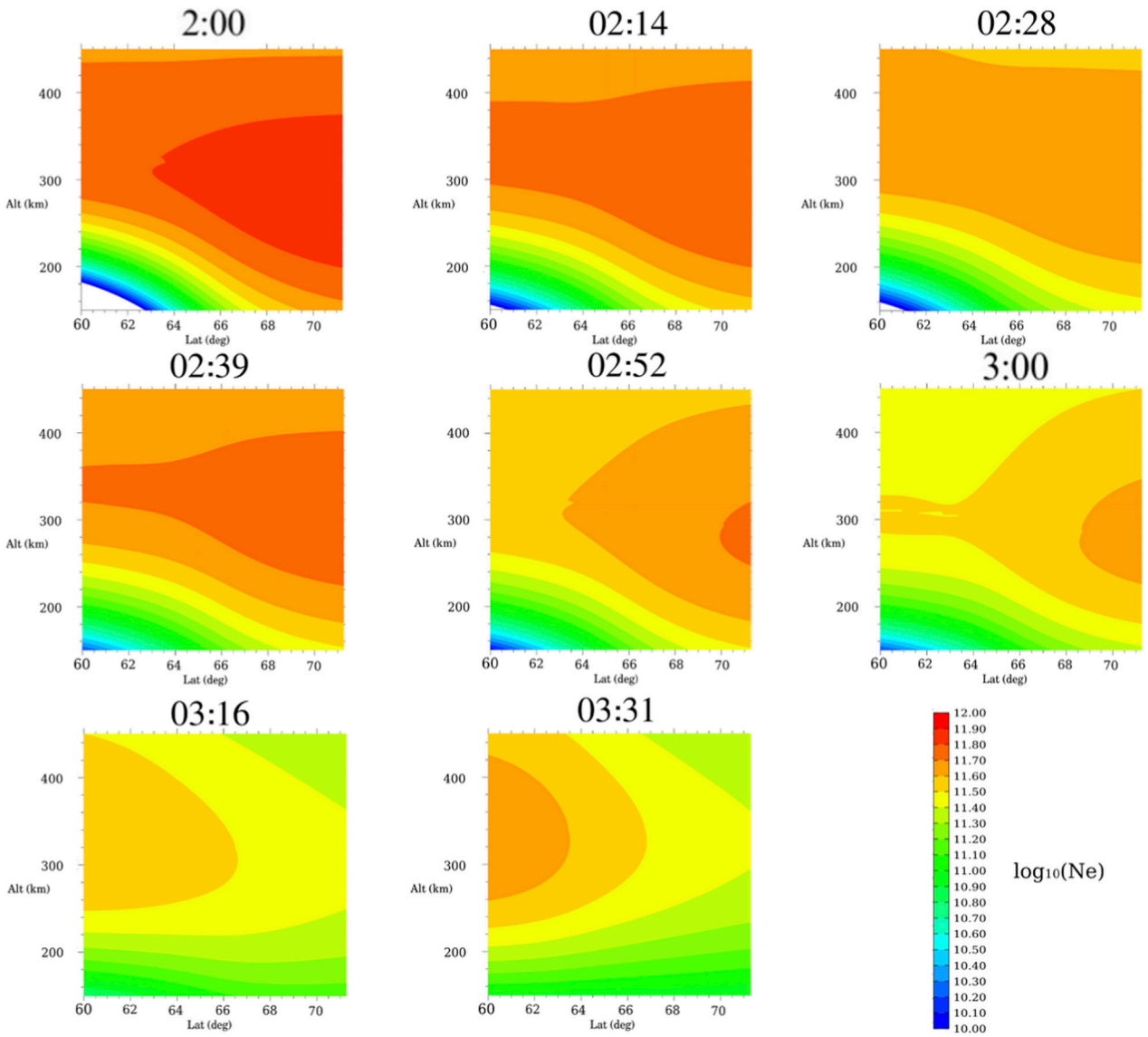


FIGURE 8
Electron density profiles from model versus latitude at different times for the fixed longitude -147°.

0240 UT) than the first half. When range spreading was present, it was always accompanied by frequency spreading, suggesting that convection is a necessary condition for irregularities and turbulence. This is unsurprising, as convection is a primary source of free energy for driving plasma instability at high latitudes. There was also less range spreading at 3.4 MHz than 2.9 MHz as a rule. There are intervals, however, when frequency spreading was unaccompanied by range spreading, notably early in the event prior to about 0210 UT. This suggests the importance of relatively small-scale irregularities at these times.

Data from the Poker Flat Incoherent Scatter Radar (PFISR) indicate that westward convection was present throughout the case-study event, strengthening drastically between the start and end and after the event. Reaching mostly velocities over 1,000 m/s at the end and after of the studied period for the latitudes between 66° and 68°. A corresponding increase in line-of-sight velocities was reported by the SuperDARN radar in Kodiak looking eastward starting at 0240 UT. This was based on the examination of velocity variations along beams 0 (west) and 15 (east) of the Kodiak radar. The increased convection detected by both instruments is consistent with the spreading increases in the HF data.

Some ray paths exhibited more range and Doppler spreading than others. This can be understood in part by considering the sensitivity to the formulas in the Supplementary Appendix to the background index of refraction. Figure 11 presents a plan view of the ray paths and the spatial distribution of the associated background indices of refraction. The index of refraction is minimized at the apex points of the ray which are approximately midway between the transmitter and receiver. The shorter the ray path, the higher the apex point in general, and the lower the background index of refraction. Figure 11 shows that the shortest (longest) ray paths had indices of refraction at their apex points as small as about 0.2 (0.7). We therefore expect the short ray paths to exhibit more broadening than the long paths and the distinction to be greater

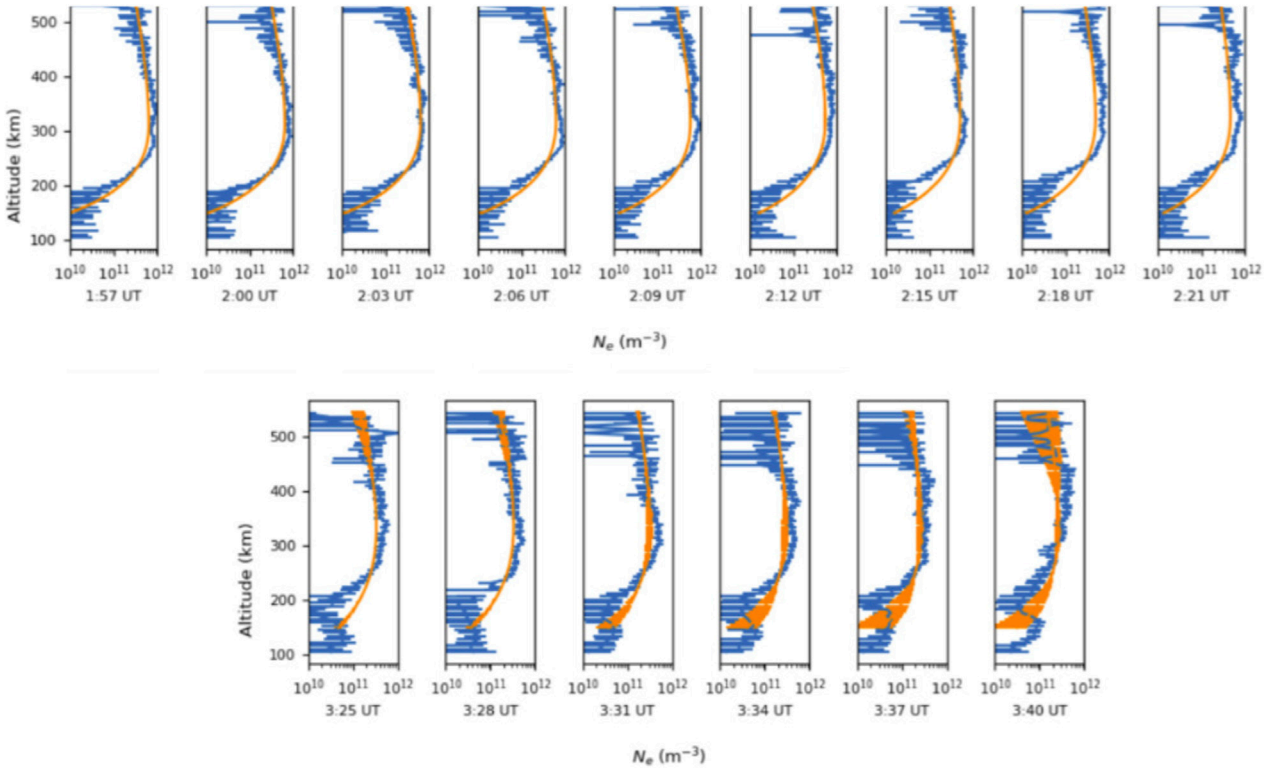


FIGURE 9
Comparison of PFISR Electron Density Profiles (in blue) with the results of the HF analysis method (in orange) including Monte Carlo Uncertainty Estimates.

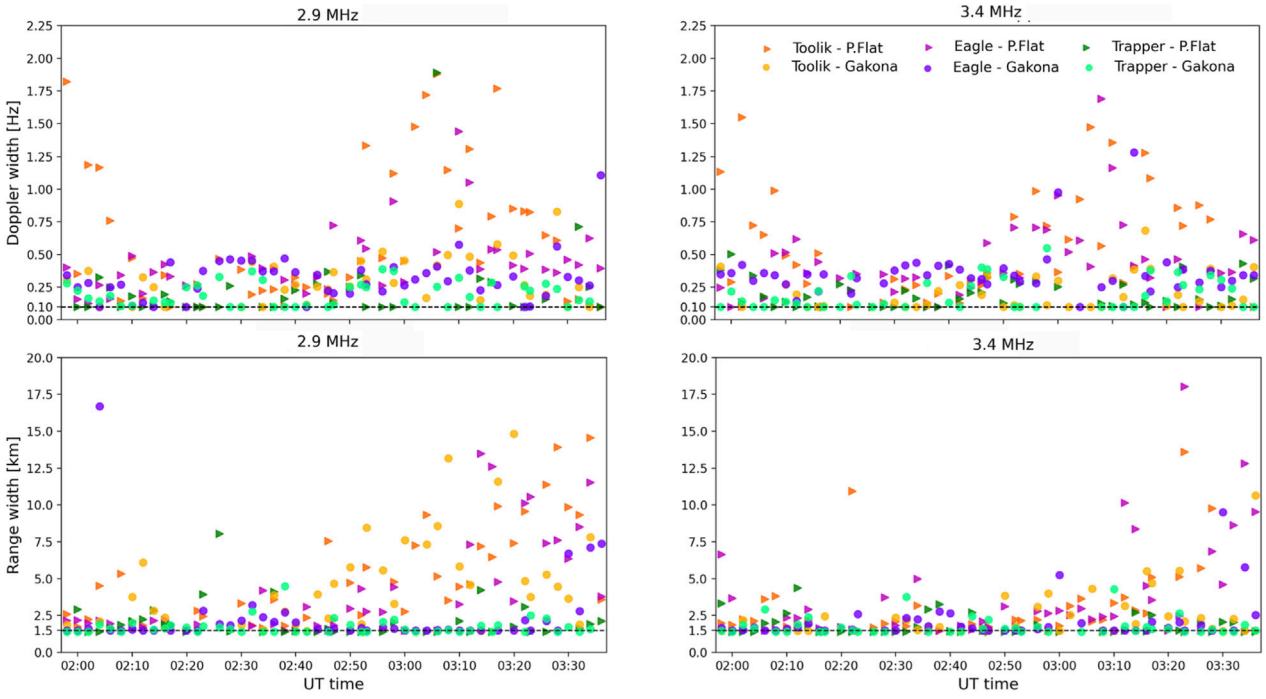


FIGURE 10
Range and Doppler spread of sky waves for different ray paths and frequencies.

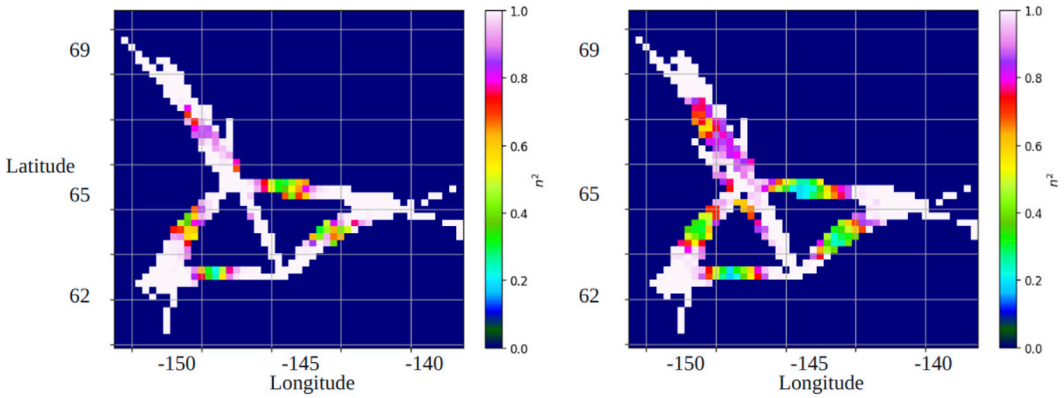


FIGURE 11
Plan view of the background index of refraction of the estimated HF ray paths at the start and end of the analysis period.

for range spreading than Doppler spreading, all other things being equal. We furthermore expect most of the spreading to occur very close to the apex points of the rays. Range spreading especially can be interpreted as a local measurement, reflecting conditions very near the apex point.

However, Figure 10 shows that the most significant range and frequency spreading occurred in Toolik ray paths followed by Eagle paths, with the Trapper Creek paths demonstrating the least spreading. The indication is that ionospheric irregularities and turbulence were most prevalent in the northernmost portions of the coverage region where background density gradients associated with the poleward trough boundary were generally steepest.

To the extent that the power spectral density in Equation 8 (Supplementary Material) is universal, it can be precomputed. In that case, the group delay, which can be converted to equivalent monostatic range by multiplication by a factor of $(c/2)^2$, is a moment of the variance of the electron number density scaled by the index of refraction to the power −6 and an isotropy factor which can be used to account for the degree to which the ray path is traversing the ionosphere along or across the magnetic field.

Following Coleman (1996), it is possible to estimate the frequency spreading at HF by populating Equation 4 with parameters characteristic of the high-latitude F region during the case study, e.g., $L_\perp = 4$ km, $L_\parallel = 50$ km, $n_\circ = 0.4$ over a path length $ds = 10$ km, $\Delta n = 10\%$ RMS, $v_\perp = 500$ m/s, and a spectral index (in Coleman's formalism) of 2.2. Doing so predicts frequency spreading of the order of 1 Hz RMS, consistent with observations. Similar calculations predict range spreading of the order of a one to a few km RMS. These results are plausible but highly sensitive to parameters which are not all well constrained.

The quantitative value of Equation 9 and Equation 8 from the Supplementary Material lies in their ratio which depends only on the carrier frequency, the $n_\circ$ floor plotted in Figure 11, the convection speed, and the spectral index to which Equation 9 is especially sensitive. The ratio of range to frequency spreading consequently represents an incisive tool for estimating the shape of the underlying, parametrized irregularity spectrum. This ratio allows us to evaluate how the width of the range changes with a specific Doppler width. Our results indicate that for a Doppler width of 1 Hz obtained with the previously stated parameters, increasing the spectral index can lead to significant spread in range. Figure 10 illustrates that range spreading can reach 10 km or more. We can achieve a spread of up to 10 km by utilizing a spectral index of 3.3 instead of 2.2. The importance of the spectral index in this ratio is highlighted by the application of solutions to the corresponding integrals of power law within the discussed ratio.

We note that range and Doppler spreading could be the result of secular changes in the group delay and Doppler shift of the HF signals occurring during the 1-min processing time, i.e., the signals could be non-stationary. The minute-to-minute consistency of the range-Doppler spectrograms together with the consistency between 10-s coherent processing intervals (CPIs) suggest stationarity, but this is not conclusive. We would argue that stationarity considerations apply to all ionospheric diagnostics during active periods, including incoherent scatter, and that the relatively high cadence of the HF measurements make the method relatively resilient by comparison to other methods.

5 Summary and discussion

A network of compact HF beacon transmitters and receivers was deployed in eastern Alaska to diagnose the F region ionosphere in the subauroral/auroral zone. The HF network performs best when absorption is low while the maximum usable frequency (MUF) is high, i.e., during quiet times in the postsunset sector. Data from the network form the foundation for volumetric electron density estimations over a broad region of space, and the inclusion of GNSS TEC and ionosonde measurements allows the reconstruction to extend into the topside. The spatial resolution depends on the density of stations in a way that is difficult to quantify. The time resolution of the measurement, meanwhile, is relatively fine for most intents and purposes.

Data from November 2023, have been used to reconstruct the shape of the main ionospheric trough. The poleward boundary, main minimum, and equatorial boundary of the trough were captured at different times of the course of less than 2 hours. Electron density estimates from the tomographic reconstruction compare favorably with overhead measurements from the Poker Flat Incoherent Scatter radar. Naturally, the limited number of HF stations available limits

the detail and fidelity of the reconstructions. As the number of ray paths from a multistatic network grows quadratically with the number of stations, however, multistatic networks have the potential to diagnose ionospheric morphology with greater fidelity than monostatic alternatives as a rule. Enlargement of the network used here is an aspirational goal.

This paper has also begun the exploration of using range and frequency spreading of the beacon signals as diagnostics of ionospheric irregularities and turbulence. Range and frequency spreading are sensitive to different parts of the turbulent wavenumber spectrum, and the latter is also sensitive to convection, although convection is a necessary condition for ionospheric instability responsible for irregularities and turbulence to begin with. While the beacon measurements reflect path-integrated quantities, we expect the broadening to reflect conditions mainly close to the apex point of a ray. In this way, the properties of the turbulence can be physically localized.

The method described here is sensitive to *D*-region ionization mainly because of nondeviative absorption and the effects on ray amplitude. However, the ability to reproduce features in the *D* region is limited and has not been explored in any depth here. In the future, we plan to combine the HF, GNSS, and ISR data with VLF measurements in order to produce a more comprehensive volumetric electron density reconstruction.

Data availability statement

The datasets presented in this study can be found in online repositories. The names of the repository/repositories and accession number(s) can be found in the article/Supplementary Material.

Author contributions

JA: Methodology, Writing – original draft, Formal Analysis, Conceptualization, Writing – review and editing. DH: Conceptualization, Project administration, Writing – review and editing, Methodology, Writing – original draft, Formal Analysis. YM: Writing – original draft, Methodology, Writing – review and editing.

Funding

The author(s) declared that financial support was received for this work and/or its publication. This work was supported by award FA9550-23-1-0164 from the Air Force Office of Scientific Research through a subcontract from the Georgia Institute of Technology and by the Defense Advanced Research Projects Agency (DARPA) through a subcontract HR00112490535 from Ohio State University.

Acknowledgments

The authors would like to thank the Geophysical Institute and the Toolik Field Station of the University of Alaska, Faribanks for assistance with our HF beacon network.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Correction note

This article has been corrected with minor changes. These changes do not impact the scientific content of the article.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fspas.2026.1848834/full#supplementary-material

References

Aa, E., Erickson, P. J., Zhang, S.-R., Zou, S., Coster, A. J., Goncharenko, L. P., et al. (2020a). A statistical study of the subauroral polarization stream over north american sector using the millstone hill incoherent scatter radar 1979–2019 measurements. *J. Geophys. Res. Space Phys.* 125. doi: 10.1029/2020JA028584

Aa, E., Zou, S., Erickson, P., Zhang, S.-R., and Liu, S. (2020b). Statistical analysis of the main ionospheric trough using swarm *in situ* measurements. *J. Geophys. Res. Space Phys.* 125. doi: 10.1029/2019JA027583

Aricoche, J. A., and Hysell, D. L. (2024). Ionospheric radio beacon signal analysis and parameter estimation using automatic differentiation. *J. Geophys. Res. Mach. Learn. Comput.* 1. doi: 10.1029/2024JH000270

Baumgarten, Y., Psiaki, M. L., and Hysell, D. L. (2021). Navigation and ionosphere characterization using high-frequency signals: models and solution concepts. *NAVIGATION* 68, 353–367. doi: 10.1002/navi.413

Bourne, H. W. (2016). *An Algorithm for Accurate Ionospheric Total Electron Content and Receiver Bias Estimation Using GPS Measurements*. Fort Collins, CO: Colorado State University.

Coleman, C. J. (1996). A model of HF sky wave radar clutter. *Radio Sci.* 31, 869–875. doi: 10.1029/96RS00721

Coleman, C. J. (2011). Point-to-point ionospheric ray tracing by a direct variational method. *Radio Sci.* 46. doi: 10.1029/2011RS004748

Coster, A., Williams, J., Weatherwax, A., Rideout, W., and Herne, D. (2013). Accuracy of gps total electron content: gps receiver bias temperature dependence. *Radio Sci.* 48, 190–196. doi: 10.1002/rds.20011

Dana, R. A., and Wittwer, L. A. (1991). A general channel model for RF propagation through structured ionization. *Radio Sci.* 26, 1059–1068. doi: 10.1029/91RS00263

Ding, M., Tong, P., Wei, Y., and Yu, L. (2021). Multiple phase screen modeling of HF wave field scintillations caused by the irregularities in inhomogeneous media. *Radio Sci.* 56, e2020RS007239. doi: 10.1029/2020RS007239.E2020RS0072392020RS007239

Fiorio, C., and Gustedt, J. (1996). Two linear time union-find strategies for image processing. *Theor. Comput. Sci.* 154, 165–181. doi: 10.1016/0304-3975(94)00262-2

Gherm, V. E., Zernov, N. N., and Strangeways, H. J. (2005). HF propagation in a wideband ionospheric fluctuating reflection channel: physically based software simulator of the channel. *Radio Sci.* 40. doi: 10.1029/2004RS003093

Hansen, P. C. (2010). *Discrete Inverse Problems: Insights and Algorithms*. Philadelphia: SIAM.

Hysell, D. L., Baumgarten, Y., Milla, M. A., Valdez, A., and Kuyeng, K. (2018). Ionospheric specification and space weather forecasting with an HF beacon network in the Peruvian sector. *J. Geophys. Res. Space Phys.* 123, 6851–6864. doi: 10.1029/2018JA025648

Hysell, D. L., Rojas, E., Goldberg, H., Milla, M. A., Kuyeng, K., Valdez, A., et al. (2021). Mapping irregularities in the postsunset equatorial ionosphere with an expanded network of HF beacons. *J. Geophys. Res. Space Phys.* 126, e2021JA029229. doi: 10.1029/2021JA029229

Jones, R., and Stephenson, J. (1975). *A Versatile three-dimensional Ray Tracing Computer Program for Radio Waves in the Ionosphere*. Washington, D.C.: U. S. Department of Commerce.

Kelley, M. C. (1989). *The Earth's Ionosphere: Plasma Physics and Electrodynamics*. Cambridge, MA: Academic Press.

Knecht, R. W., and Zandt, T. E. V. (1963). "Some early results from the ionospheric top-side sounder satellite,". Boulder, Colorado.*Tech. Rep*.

Morozov, V. A. (1966). On the solution of functional equations by the method of regularization. *Sov. Math. Dokl.* 7, 414–417.

Morton, Y. J., Yang, Z., Breitsch, B., Bourne, H., and Rino, C. (2020). Ionospheric effects, monitoring, and mitigation techniques 1, 879–937.

Moses, W. S., Churavy, V., Paehler, L., Hückelheim, J., Narayanan, S. H. K., Schanen, M., et al. (2021). "Reverse-mode automatic differentiation and optimization of GPU kernels via enzyme," in *SC21: International Conference for High Performance Computing, Networking, Storage and Analysis*, 1–18. doi: 10.1145/3458817.3476165

Moses, W. S., Narayanan, S. H. K., Paehler, L., Churavy, V., Schanen, M., Hückelheim, J., et al. (2022). "Scalable automatic differentiation of multiple parallel paradigms through compiler augmentation," in *SC22: International Conference for High Performance Computing, Networking, Storage and Analysis*, 1–18. doi: 10.1109/SC41404.2022.00065

Muldrew, D. B. (1965). F-layer ionization troughs deduced from alouette data. *J. Geophys. Res.* 70, 2635–2650. doi: 10.1029/JZ070i011p02635

Nickisch, L. J., St. John, G., Fridman, S. V., Hausman, M. A., and Coleman, C. J. (2012). HiCIRF: a high-fidelity HF channel simulation. *Radio Sci.* 47. doi: 10.1029/2011RS004928

Powell, M. J. D. (1964). An efficient method for finding the minimum of a function of several variables without calculating derivatives. *Comput. J.* 7, 155–162. doi: 10.1093/comjnl/7.2.155.Hdl:10338.dmlcz/103029

Quegan, S., Rodger, A. S., Williams, P. J. S., Lockwood, M., and Rishbeth, H. (1989). The influence of convection on the structure of the high-latitude ionosphere [and discussion]. *Philosophical Trans. R. Soc. Lond. Ser. A, Math. Phys. Sci.* 328, 119–137.

Ravan, M., Riddolls, R. J., and Adve, R. S. (2012). Ionospheric and auroral clutter models for HF surface wave and over-the-horizon radar systems. *Radio Sci.* 47. doi: 10.1029/2011RS004944

Rodger, A. (2008). *The Mid-latitude Trough—Revisited*. American Geophysical Union, 25–33. doi: 10.1029/181GM04

Schunk, R. W., and Nagy, A. F. (2004). *Ionospheres: Physics, Plasma Physics, and Chemistry*. Cambridge, UK: Cambridge University Press.

Thébault, E., Finlay, C., Beggan, C., Alken, P., Aubert, J., Barrios, O., et al. (2015). International geomagnetic reference field: the 12th generation. *Earth, Planets Space* 67. doi: 10.1186/s40623-015-0228-9

Tineo, G. (2026). "Development of an sdr ionosonde using red pitaya signallab 250-12 board," in *International Symposium on Equatorial Aeronomy (Liberia, Costa Rica)*.

Vierinen, J., Chau, J. L., Pfeffer, N., Clahsen, M., and Stober, G. (2016). Coded continuous wave meteor radar. *Atmos. Meas. Tech.* 9, 829–839. doi: 10.5194/amt-9-829-2016

Wagen, J.-F., and Yeh, K. C. (1989). Simulation of HF propagation and angle of arrival in a turbulent ionosphere. *Radio Sci.* 24, 196–208. doi: 10.1029/RS024i002p00196

Wagner, L. S., and Goldstein, J. A. (1985). High-resolution probing of the hf ionospheric skywave channel: F2 layer results. *Radio Sci.* 20, 287–302.

Wu, K., Otoo, E., and Shoshani, A. (2005). Optimizing connected component labeling algorithms. *Tech. Rep*.

Zernov, N. N., Gherm, V., and Zaalov, N. Y. (2019). "High-frequency EM wave field propagation in the disturbed ionosphere: review of recent research at St. Petersburg state university," in *Proceedings of the 2019 Russian Open Conference on Radio Wave Propagation (Kazan, Russia)*, 25–30.